

Spherical Tammes Distribution for 24 Points

1 Problem statement and objective function

Problem 34 requires placing points on the unit sphere so that the distance in the closest pair is as large as possible. I chose the fixed case $N = 24$. A GigaEvo problem requires one output format, and a solution is known for this value of N , so the result can be compared with the proven value [1].

Let

$$\mathbb{S}^2 = \{x \in \mathbb{R}^3 : \|x\|_2 = 1\}, \quad P = (p_1, \dots, p_{24}) \in (\mathbb{S}^2)^{24}. \quad (1)$$

For a configuration P , the score is

$$\delta(P) = \min_{1 \leq i < j \leq 24} \|p_i - p_j\|_2, \quad C(24) = \max_{P \in (\mathbb{S}^2)^{24}} \delta(P). \quad (2)$$

There are

$$\binom{24}{2} = 276$$

unordered pairs in total. The function `entrypoint()` returns a real array of shape $(24, 3)$, with one point written in each row. Coincident rows are allowed.

This formalization agrees with the original problem: the sphere is the unit sphere, Euclidean chord distance is used, the minimum is taken over all pairs, and then this minimum is maximized. An optimal configuration exists. Indeed, the space $(\mathbb{S}^2)^{24}$ is compact, each pairwise distance is continuous, and the minimum of a finite number of continuous functions is also continuous.

There are two convenient equivalent forms. For unit vectors,

$$\|p_i - p_j\|_2^2 = \|p_i\|_2^2 + \|p_j\|_2^2 - 2p_i \cdot p_j, \quad (3)$$

$$= 2 - 2p_i \cdot p_j. \quad (4)$$

Let

$$\mu(P) = \max_{1 \leq i < j \leq 24} p_i \cdot p_j,$$

and let $\theta(P)$ denote the smallest central angle between the points. Then

$$\delta(P) = \sqrt{2 - 2\mu(P)} = 2 \sin \left(\frac{\theta(P)}{2} \right). \quad (5)$$

Therefore, maximizing the smallest distance is equivalent to minimizing the largest inner product or maximizing the smallest angle.

2 What is known about the answer

First, one can obtain the correct order of magnitude for arbitrary N . A spherical cap of angular radius r on the unit sphere has area

$$A(r) = 2\pi(1 - \cos r). \quad (6)$$

If the smallest angular distance is θ , then the caps of radius $\theta/2$ around N points have disjoint interiors. Therefore,

$$N 2\pi \left(1 - \cos \frac{\theta}{2} \right) \leq 4\pi. \quad (7)$$

Hence,

$$\cos \frac{\theta}{2} \geq 1 - \frac{2}{N}.$$

Substituting

$$C(N) = 2 \sin \frac{\theta}{2},$$

we obtain the upper bound

$$C(N) \leq \frac{4\sqrt{N-1}}{N}. \quad (8)$$

For the lower bound, the points can be chosen greedily. If fewer than N points have been chosen and

$$(N-1)A(\theta) < 4\pi, \quad (9)$$

then the caps of radius θ around the already chosen points do not cover the whole sphere. Therefore, one more point can be added outside their union. Passing to the limiting value of θ , we obtain

$$C(N) \geq \frac{2}{\sqrt{N-1}}. \quad (10)$$

Thus,

$$\frac{2}{\sqrt{N-1}} \leq C(N) \leq \frac{4\sqrt{N-1}}{N}, \quad (11)$$

so $C(N)$ has order $N^{-1/2}$. For $N = 24$, the elementary bounds are

$$0.4170288 \leq C(24) \leq 0.7993053.$$

This is only a scale check, not an exact solution.

The exact case $N = 24$ was solved by Robinson. The optimal configuration consists of the 24 vertices of a snub cube normalized to the unit sphere. (We refer to R. M. Robinson, *Arrangement of 24 points on a sphere.*) In the table of spherical codes [2], the inner product c of the closest pair is given as the real root of the equation

$$7c^3 + c^2 - 3c - 1 = 0, \quad c \approx 0.7230784683335085. \quad (12)$$

Passing to the chord distance computed by the validator, we obtain

$$C(24) = \sqrt{2-2c} \approx 0.7442063311562076. \quad (13)$$

3 Baseline and improved construction

The simple baseline is constructed using a Fibonacci spiral. For $i = 0, \dots, 23$, the formulas are

$$z_i = 1 - \frac{2(i+1/2)}{24}, \quad (14)$$

$$\varphi_i = \pi(1 + \sqrt{5})(i+1/2), \quad (15)$$

$$p_i = \left(\sqrt{1-z_i^2} \cos \varphi_i, \sqrt{1-z_i^2} \sin \varphi_i, z_i \right). \quad (16)$$

In the improved baseline, the same Fibonacci sphere is constructed first, after which the function

$$E_s(P) = \sum_{i < j} \|p_i - p_j\|_2^{-s}$$

is minimized.

Let the minimum distance between the points be

$$\delta(P) = \min_{i < j} \|p_i - p_j\|_2.$$

For 24 points, there are

$$M = \binom{24}{2} = 276$$

pairs. At least one term in E_s is equal to $\delta(P)^{-s}$, and the others do not exceed it. Therefore,

$$\delta(P)^{-s} \leq E_s(P) \leq M\delta(P)^{-s}.$$

After raising to the power $-1/s$, we obtain

$$M^{-1/s}\delta(P) \leq E_s(P)^{-1/s} \leq \delta(P).$$

As s increases, the value $M^{-1/s}$ tends to one. Consequently, $E_s(P)^{-1/s}$ approaches the minimum distance $\delta(P)$. Therefore, minimizing E_s is a justified approximation to the original problem of maximizing the minimum distance.

The algorithm starts with $s = 1$ and then gradually increases it. For small values of s , the arrangement of all points is taken into account, while for large values the closest pairs have the main influence. After each step, the points are normalized, so they remain on the unit sphere.

The following scores were obtained:

Construction	Fitness	Absolute gap	Relative gap
Fibonacci sphere	0.6300755024	0.1141308288	15.335912%
Riesz continuation	0.7441981543	0.0000081768	0.001099%

The improved result is very close to the theoretical value, but is not equal to it. A possible continuation of the work is to refine the numerical coordinates of the obtained construction and investigate the pairs of points at which the minimum distance is attained.

4 Edge cases and numerical precision

First, the validator requires the exact shape (24, 3). It rejects non-numeric objects, complex values, `NaN`, infinities, and values that cannot be safely represented in `float64`. Norms are computed using nested calls to `hypot`, so even very large finite coordinates are rejected correctly rather than causing overflow.

Each row must satisfy the condition

$$|||p_i||_2 - 1| \leq \tau, \quad \tau = 10^{-7}. \quad (17)$$

After this, the accepted rows are normalized. This normalization prevents the obvious attempt to increase distances by radial scaling. If

$$\hat{p} = \frac{p}{||p||_2},$$

then

$$||p - \hat{p}||_2 = |||p||_2 - 1| \leq \tau. \quad (18)$$

For two accepted points, the triangle inequality implies

$$|||p_i - p_j||_2 - ||\hat{p}_i - \hat{p}_j||_2| \leq 2\tau. \quad (19)$$

Therefore, the tolerance has a clear geometric meaning, and the final score is computed for points lying exactly on the sphere.

All 276 distances are computed from coordinate differences. For nearly coincident points, this is more accurate than the formula $2 - 2p_i \cdot p_j$, in which two close numbers are subtracted. Distances from points to themselves are excluded. Exact duplicates are not rejected: they are valid and correctly receive fitness = 0. Clusters and nearly coincident points similarly receive a small score.

For valid output, the function `validate()` returns `fitness` and `is_valid = 1`. Invalid data raise `ValueError`, after which the metric specification assigns the values `fitness = -1` and `is_valid = 0` for GigaEvo.

A set of 22 tests checks both constructions, determinism, invariance under rotation and permutation, the tolerance boundary, duplicates, incorrect shapes, complex and non-numeric data, `NaN`, infinities, huge finite values, and immutability of the input array.

References

- [1] Google DeepMind. Problem 34: Tammes problem. https://google-deepmind.github.io/alphaevolve_repository_of_problems/problems/34.html
- [2] H. Cohn. Table of spherical codes. <https://cohn.mit.edu/spherical-codes/>